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Theoretical Studies of Enzymic Reactions:

Dielectric, Electrostatic and Steric Stabilization of the Carbonium Ion in the Reaction of Lysozyme

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A general method for detailed study of enzymic reactions is presented. The method considers the complete enzyme–substrate complex together with the surrounding solvent and evaluates all the different quantum mechanical and classical energy factors that can affect the reaction pathway. These factors include the quantum mechanical energies associated with bond cleavage and charge redistribution of the substrate and the classical energies of steric and electrostatic interactions between the substrate and the enzyme. The electrostatic polarization of the enzyme atoms and the orientation of the dipoles of the surrounding water molecules is simulated by a microscopic dielectric model. The solvation energy resulting from this polarization is considerable and must be included in any realistic calculation of chemical reactions involving anything more than an isolated molecule *in vacuo*. Without it, acidic groups can never become ionized and the charge distribution on the substrate will not be reasonable. The same dielectric model can also be used to study the reaction of the substrate in solution. In this way the reaction in solution can be compared with the enzymic reaction.

In this paper we study the stability of the carbonium ion intermediate formed in the cleavage of a glycosidic bond by lysozyme. It is found that electrostatic stabilization is an important factor in increasing the rate of the reaction step that leads to the formation of the carbonium ion intermediate. Steric factors, such as the strain of the substrate on binding to lysozyme, do not seem to contribute significantly.

1. Introduction

Understanding the detailed mechanism of enzymic catalysis is one of the fundamental problems in biochemistry. Powerful X-ray crystallographic and biochemical methods have provided very many details about the mechanism of some enzymic reactions (see *Cold Spring Harbor Symp. Quant. Biol.* 1972) but a complete and general understanding of these reactions is still to be achieved. At the present time, the exact role, relative importance, and generality of such factors as strain (Phillips, 1966; Blake *et al.*, 1967), acid catalysis, charge stabilization (Vernon, 1967), charge relay systems (Blow *et al.*, 1969), orbital steering (Storm & Koshland, 1970) and entropy (Page & Jencks, 1971) are still not clear. Estimation of the relative importance of all the factors involved in enzymic reactions requires a proper theoretical method that can be used together with the available experimental information. The development and examination of such a theoretical scheme is the main purpose of this paper.

The conformation of the enzyme–substrate complex can be adequately studied using empirical energy functions based on the "classical" contributions of bond stretching, bond angle bending, bond twisting, non-bonded interactions, etc. (Levitt, 1974*a*; Platzer *et al.*, 1972). The mechanism and energetics of the enzymic reaction can only be studied using a quantum mechanical approach. Previous quantum mechanical calculations on enzymic reactions have been limited in several respects. In the first place, they deal with an over-simplified model system consisting of only small fractions of the atoms involved in the real enzyme–substrate interaction (Loew & Thomas, 1972; Umeyama *et al.*, 1973; Scheiner *et al.*, 1975). Second, all available quantum mechanical methods treat the reaction as if in a vacuum and are, therefore, unable to include the dielectric, which has a very important effect on the energy and charge distribution of the system. This limitation does not depend on the actual quantum mechanical scheme, being equally valid for the simplest Hückel treatment and most extensive *ab initio* calculation.

Because the treatment of the whole enzyme–substrate complex quantum mechanically is computationally impossible, it is necessary to simplify the problem and use a hybrid classical/quantum mechanical approach. The method described here considers the whole enzyme–substrate complex: the energy and charge distribution of those atoms that are directly involved in the reaction are evaluated quantum mechanically, while the potential energy surfac e of the rest of the system, including the urrounding solvent, is evaluated classically. An important feature of the method is the treatment of the "dielectric" effect due to both the protein atoms and the surrounding water molecules. Our dielectric model is based on a direct calculation of the electrostatic field due to the dipoles induced by polarizing the protein atoms, and the dipoles induced by orienting the surrounding water molecules. With the inclusion of the microscopic dielectric effect we feel that our model for enzymic reactions includes all the important factors that may contribute to the potential surface of the enzymic reaction. Because we use analytical expressions to evaluate the energy and first derivative with respect to the atomic Cartesian co-ordinates, it is possible to allow the system to relax using a convergent energy minimization method.

Here we use the method to study the mechanism of the lysozyme reaction, with particular emphasis on factors contributing to the stabilization of the carbonium ion intermediate. We first outline our theoretical scheme, paying attention to the partition of the potential surface into classical and quantum mechanical parts and to details of the microscopic dielectric model. Then we use the method to investigate factors that contribute to the stabilization of the carbonium ion intermediate and deal with the relative importance of each factor. The effect of steric strain is considered for both the ground and the carbonium states and found to be of minor importance. Electrostatic stabilization is treated at some length with special emphasis on the balance between stabilization by induced dipoles (polarizability) and stabilization by charge–charge interactions. The results demonstrate the importance of electrostatic stabilization for this reaction.

2. Methods

Available quantum mechanical treatments of chemical reactions are valid only so long as they deal with an isolated system in vacuum; when a part of a complete system is treated separately, the results have limited validity; for example, if only the reactants are treated in the case of an ionic reaction in solution. The theoretical scheme presented here overcomes this problem by considering the complete system explicitly. We use a hybrid classical/quantum mechanical approach that restricts the quantum mechanical treatment to those atoms of the enzyme–substrate complex that are directly involved in the reaction. The energy and charge distribution of the non-quantum mechanical part of the system are described by classical potential functions and a classical charge distribution. The classical and the quantum mechanical parts of the energy are coupled by allowing for the changes in the quantum mechanical charges due to the electrostatic potential from charges and induced dipoles in the classical region, and by considering the electrostatic and van der Waals interactions between the atoms in the quantum and the classical regions. The "dielectric" effect due to the polarization of the enzyme atoms and the orientation of the water dipole moments is included in the calculation and fully described in section (b), below.

(a) *Partition of the potential energy*

The potential energy surface of the whole system is decomposed into the following terms:

$$V = V_{\text{classical}} + V_{\text{quantum}} + V_{\text{quantum/classical}}, \tag{1}$$

where the first two terms are the classical and the quantum mechanical potential energies, respectively, and the third is the coupling term. The partition of the lysozyme/hexa N-acetyl glucosamine (NAG) complex into quantum and classical regions is shown in Fig. 1. The shaded area is the quantum region; the rest of the system, including the whole protein and surrounding solvent, is the classical region.

(i) *The classical potential energy*

The classical potential energy is represented by empirical potential functions of the following form:

$$V_{\text{classical}} = \sum_i K_b(b_i - b_0)^2 + \sum_i K_\theta(\theta_i - \theta_0)^2 + \sum_i K_\theta \cos\{n(\phi_i - \phi_0)\}$$
$$+ \sum_{i>j} \epsilon_{ij}\{(r^0_{ij}/r_{ij})^{12} - 2(r^0_{ij}/r_{ij})^6\} + \sum_{i>j} Q_iQ_j/r_{ij}, \tag{2}$$

where b_i, θ_i, ϕ_i and r_{ij} are, respectively, the bond lengths, bond angles, torsional angles and interatomic distances in the whole enzyme–substrate complex. The internal co-ordinate parameters K_b, b_0, K_θ, θ_0, K_φ, n, and ϕ_0, which were determined from spectroscopic information, are presented elsewhere (Levitt, 1974*a*). The non-bonded parameters ϵ_{ij} and r^0_{ij} and the atomic partial charges Q_i were determined to fit the observed properties of hydrocarbon, amide and amino acid crystals (Levitt & Lifson, unpublished work). ϵ_{ii} = 0·0186, 0·0519, 0·1459, 0·1366, 0·0679, 0·4519 kcal/mol, respectively, and r^0_{ii} = 2·8152, 3·4617, 3·6665, 4·0866, 4·2388, 3·6343 Å, respectively, for interactions between two identical atoms of the following types: H(hydrocarbon), O(carbonyl), O′(carboxyl), N, C, C′(carbonyl and carboxyl). No van der Waals interaction is calculated for hydrogens bonded to O or N. Interactions between different atoms i, j use the geometric means of the ϵ and r^0 values given above. The atomic partial charges Q_i in fractions of an electron are 0·427, 0·253, 0·417, 0·533, 0·097, 0·097, 0·097, −0·392, −0·880, respectively, for the second atom in the following groups: $-OH$, $>NH$, $-NH_2$, $-NH_3$, $-CH$ and $>CH$, $>CH_2$, $-CH_3$, $>C'O$, $-C'O'O'$. The charge on the first atom is found by requiring group neutrality for all groups except $-NH_3$ and $-C'O'O'$, which carry a net charge of 1 and −1, respectively. All hydrogen atoms are included in the calculation. Van der Waals and electrostatic interactions were not calculated between atoms separated along the covalent structure by three or less bonds. This empirical potential function approach is simple and efficient (in terms of the computer time needed to evaluate the energy and its derivatives), yet provides a reliable approximation to the molecular potential surface. In fact, when

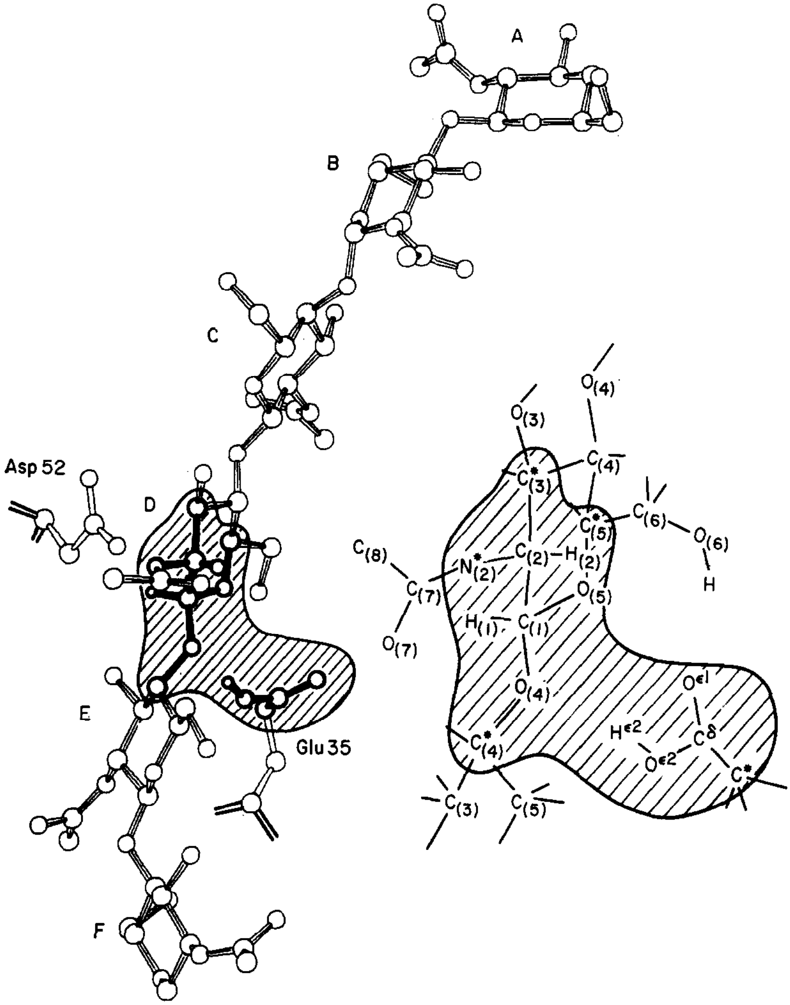

FIG. 1. Partition of the lysozyme/substrate system into quantum mechanical and classical regions. The shaded area defines the quantum mechanical region, which is presented in greater detail on the right-hand side of the Figure. Only part of the enzyme is drawn, although the whole enzyme is included in the calculation.

dealing with those regions of the ground state potential energy surfaces for which the localized electron description is adequate, the empirical approach is probably more reliable than any semi-empirical quantum mechanical approach, particularly if relaxation of steric repulsion is allowed.

(ii) *The quantum mechanical potential energy*

The quantum mechanical potential energy is obtained by extending the QCFF/PI method (Warshel & Karplus, 1972; Warshel, 1973) to include all the valence electrons (Warshel, unpublished work). The method, referred to as QCFF/ALL, is a semi-empirical quantum mechanical approach. It uses hybrid atomic orbitals with Löwdin corrections for overlap between adjacent orbitals on the same atom which are not always orthogonal in this representation. The semi-empirical integrals between these hybrid orbitals possess a clearer physical meaning than in other semi-empirical quantum mechanical methods, which use orbitals along the x,y,z axes. The quantum mechanical potential surface can be refined very conveniently as it is possible, for example, to fit the integrals between the two bonding hybrid orbitals directly to the known properties of that type of bond. Another important feature of hybrid orbitals is the representation of a covalent connection between the quantum and the classical parts of the molecule by a single orbital. The parameterization of the method was done as before (Lifson & Warshel, 1968; Warshel & Lifson, 1970; Warshel & Karplus, 1972) by fitting of a large set of calculated and observed molecular properties. The QCFF/ALL method provides an analytical first derivative of the quantum mechanical potential energy which is of great importance for efficient energy minimization. As the quantum mechanical energies of bond breaking are overestimated by methods which use only a single Slater determinant, we included the double excitation Configuration Interaction corrections in the calculation (Daudel *et al.*, 1959).

In order to estimate the sensitivity of our results to the parameterization, we also used the more conventional MINDO/2 semiempirical molecular orbital method (Dewar & Haselbach, 1970). When using the MINDO/2 method we had to introduce an extra geometrical constraint on the bond angles of atoms treated quantum mechanically, as the use of a single orbital in the covalent connections between the quantum and classical regions can lead to instability for the unhybridized MINDO orbitals.

(iii) *Classical/quantum mechanical coupling*

Although the quantum mechanical part of the calculation gives the net atomic charges of the isolated quantum system, it is important to include the inductive effect of the classical point charges on the quantum charges. Writing the diagonal SCF (self-consistent field) matrix element $F^{ii}_{\mu\mu}$ as if the whole complex were to be treated quantum mechanically in the zero differential overlap approximation (see for example, Pople & Beveridge, 1970), one obtains

$$F^{ii}_{\mu\mu} = U_{\mu\mu} + \tfrac{1}{2}P_{\mu\mu}\gamma_{\mu\mu} - \sum_{\nu\neq\mu}^{i} P_{\nu\nu}\gamma_{\nu\mu} - \sum_{i'\neq i} Q_{i'}\gamma_{i'} - \sum_{j} Q_j\gamma_{ij}, \tag{3}$$

where U is the core Hamiltonian, P is the quantum mechanical bond order, Q is the net atomic charge, and γ is the repulsive integral, while μ and ν are atomic orbitals on atom i. Atom i is any particular atom in the quantum mechanical region, the index i' runs over all other atoms in that region and the index j runs over all the other atoms in the classical region. Assuming that the charge density on atom j is constant (as implied by not including its orbitals in the SCF iterations), and replacing γ_{ij} by $1/r_{ij}$ gives

$$F^{ii}_{\mu\mu} = \bar{F}^{ii}_{\mu\mu} - \sum_{j} Q_j/r_{ij} + (\Delta F^{ii}_{\mu\mu})_{\text{ind}}, \tag{4}$$

where $\bar{F}^{ii}_{\mu\mu}$ is the corresponding matrix element for the isolated quantum mechanical part (atoms i only), and the evaluation of $(\Delta F^{ii}_{\mu\mu})_{\text{ind}}$ is described in the next section. The energy contribution due to the external partial charge Q_j is given simply by the electrostatic interaction between atoms i and j

$$V_{QQ} = \sum_{i,j} Q_iQ_j/r_{ij}. \tag{5}$$

With this contribution, the potential that couples the quantum and the classical surfaces is given by:

$$V_{\text{quantum/classical}} = \sum_{i,j} Q_iQ_j/r_{ij} + \sum_{i,j} \epsilon_{ij}\{(r^0_{ij}/r_{ij})^{12} - 2(r^0_{ij}/r_{ij})^6\} + V^{\text{E}}_{\text{ind}} + V^{\text{W}}_{\text{ind}}$$
$$(i \text{ in quantum } j \text{ in classical}). \tag{6}$$

The second term accounts for the van der Waals interactions, which are treated classically. The covalent bonds between quantum and classical atoms are treated classically. The quantum atom in such a bond is represented by only a single hybrid orbital (see Fig. 2). The evaluation of $V^{\text{E}}_{\text{ind}}$ and $V^{\text{W}}_{\text{ind}}$ is described in the next section.

(b) *Induced dipoles and the effective dielectric*

Net charges, like those that are formed when an acidic group is ionized, cause very strong electric fields over an extensive region of space. When this region is not vacuum, the atoms in it will be polarized, giving induced dipoles which interact with the original

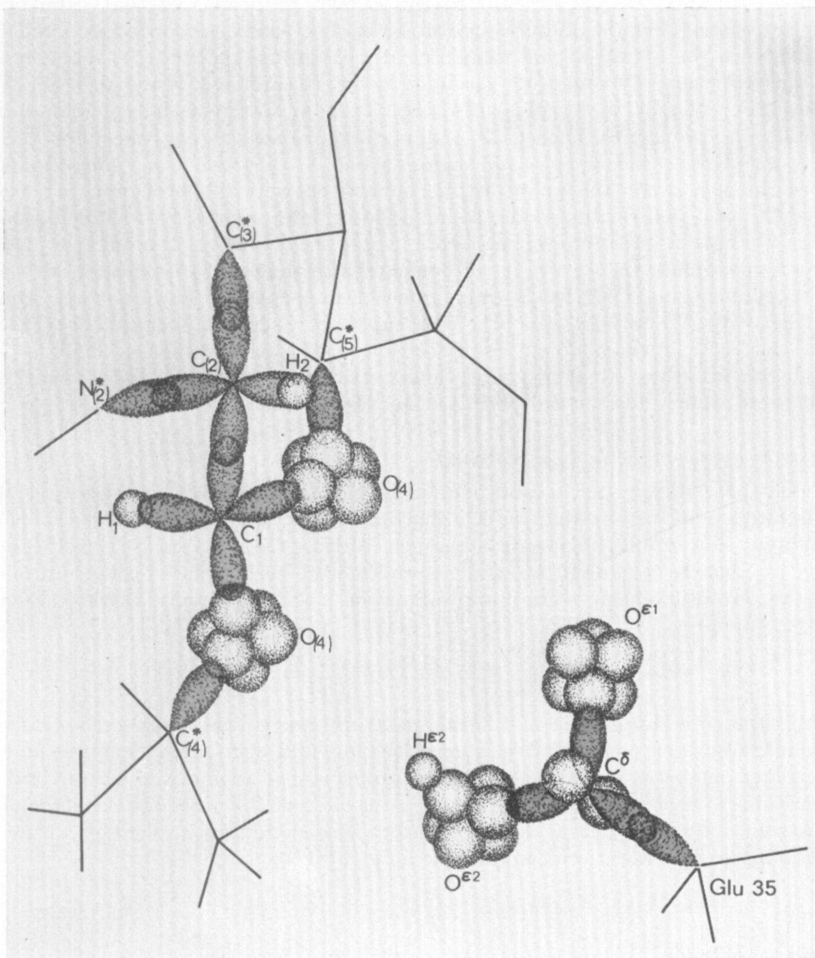

FIG. 2. The hybrid atomic orbitals that are included in the quantum mechanical region of the lysozyme system. The 2*s* orbitals of the oxygen atoms which are included in the calculation are not drawn in the Figure. Note the single sp^2 orbital on C* and N* at the boundaries of the quantum mechanical and classical regions.

electrostatic field of the net charge. This interaction always reduces the energy of the system as the induced dipoles are always parallel to the electric field inducing them. If this polarization is not included, then the energy of the system rises considerably when groups are ionized and charges of opposite sign are separated.

Here we allow for stabilization of charges in three ways: (1) by interaction with other atomic partial charges in the vicinity (this is just part of the formal charge–charge interaction in eqns (2) and (5); (2) by interaction with dipoles induced on the atoms; and (3) by interaction with the permanent dipoles of the surrounding water molecules that are oriented by the electric field. Because the interaction with induced atomic dipoles and the solvent molecules depends on the shape of the active site, we did not try to use various theories for the dielectric effect inside a cavity of simple shape (Onsager, 1936; Kirkwood & Westheimer, 1938). Instead, we developed the following general model for the treatment of the microscopic dielectric: the system is divided into 3 regions (see Fig. 3). (I) The quantum mechanical atoms of the substrate and the enzyme and those classical atoms of the enzyme close to the active site that are assumed to be ionized. (II) All other atoms of

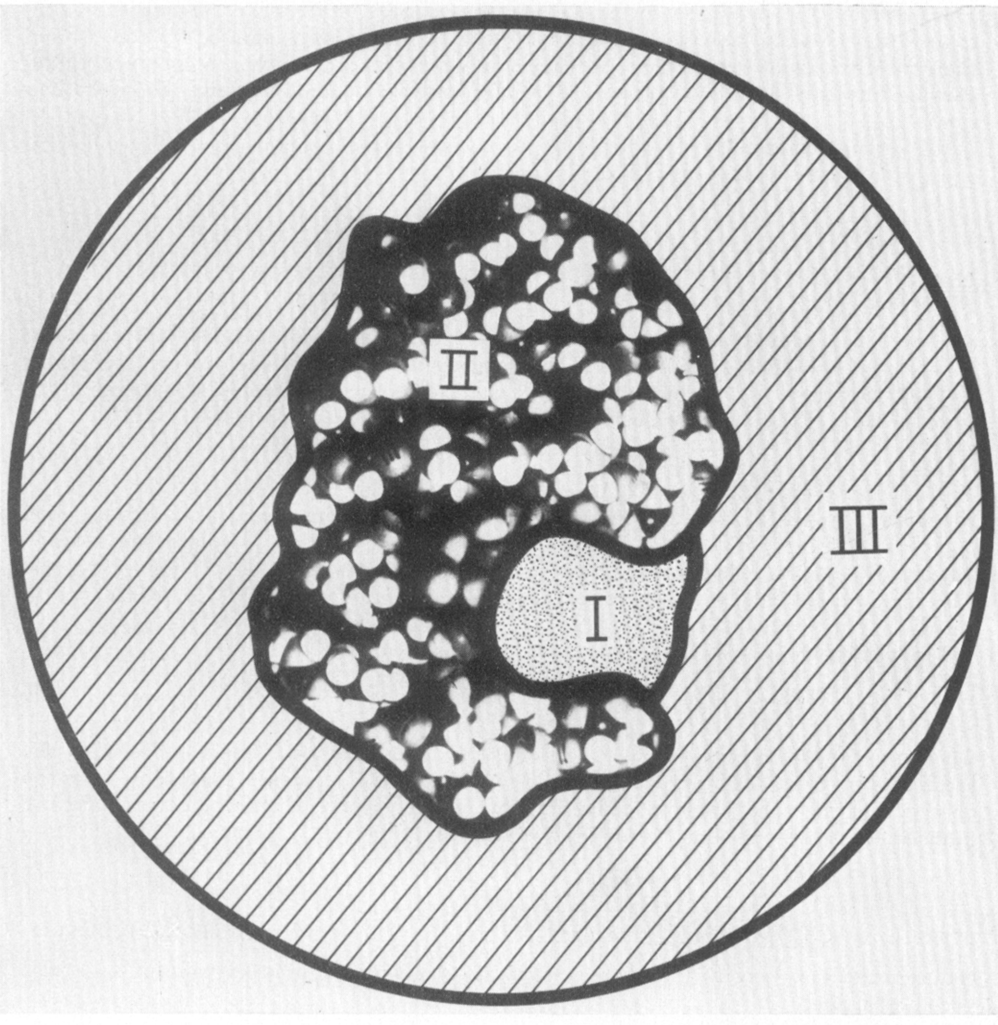

FIG. 3. Definition of the regions used in the microscopic dielectric model. Regions (I), (II) and (III) contain quantum mechanical atoms and the ionized groups, the polarized atoms of the enzyme, and the surrounding water molecules, respectively. The picture of the space-filling model of lyzosyme is taken from Dickerson & Geis (1969).

the protein and the non-quantum mechanical atoms of the substrate. (III) The water molecules around the protein.

In regions I and II the dielectric effect is due to the polarization of individual atoms leading to induced dipoles. In region I this inductive effect is produced automatically by the quantum mechanical calculation. In region II we assign a point-induced dipole $\boldsymbol{\mu}$ to each atom by the relation

$$\boldsymbol{\mu}_j = \alpha_j \mathbf{E}_j, \tag{7}$$

where α is the atomic polarizability, and the local electric field on atom j, $\mathbf{E}_j$, is given by the standard expression:

$$\mathbf{E}_j = \sum_i Q_i \mathbf{r}_{ij}/r_{ij}^3 + \sum_{j' \neq j} - \nabla(\mu_{j'} \mathbf{r}_{j'j}/r_{j'j}^3). \tag{8}$$

It is possible to solve eqn (8) for $\mathbf{E}_j$ and eqn (7) for $\boldsymbol{\mu}_j$ by an iterative self-consistent procedure: the initial $\boldsymbol{\mu}_j$ values are set to zero, the $\mathbf{E}_j$ values are evaluated by eqn (8) and then substituted into eqn (7) to provide new values for $\boldsymbol{\mu}_j$, which are once again substituted into eqn (8). Such a procedure usually converges completely after about 7 iterations.

For greater computational efficiency, we choose to replace eqn (8) by the following expression for the local electric field

$$\mathbf{E}_j = \sum_i Q_i \mathbf{r}_{ij} / \{ r_{ij}^3 \, \epsilon(r) \}, \tag{9}$$

where $\epsilon(r)$ has the same functional form for any i,j pair and accounts for the reduction of the field due to the central charges, $\sum_i Q_i \mathbf{r}_{ij}/r_{ij}^3$, by the back field from the surrounding dipoles. This attenuation factor, which is related to the conventional dielectric constant, is only used to calculate the local field that induces the dipoles on the surrounding atoms, and not to reduce the interaction between charges. $\epsilon(r)$ was estimated as a function of r from the ratio of the actual self-consistent field (eqns (7) and (8)) to the *in vacuo* field

$$\epsilon(r) = \{\sum |\mathbf{E}_j|^2\}^{\frac{1}{2}} / \{\sum_j |Q\mathbf{r}_j/r_j^3|^2\}^{\frac{1}{2}}, \tag{10}$$

where the j atoms in the summation are at distances r_j from a central charge, Q, with $r \leq r_j < r + \Delta r$ and $\Delta r = 0{\cdot}5$ Å. The polarizability, α, used to calculate the induced dipoles (eqn (7)) is taken as 1 Å^3 for non-hydrogen atoms and 0·5 Å^3 for hydrogens. Although $\epsilon(r)$ did vary with separation r, with a maximum at about 4·5 Å, we decided to use the average value of 1·36.

The energy of this system of charges and dipoles is calculated as

$$V_{\mathrm{ind}}^{\mathrm{E}} = - \sum_{i,j} Q_i (\boldsymbol{\mu}_j \cdot \mathbf{r}_{ij})/r_{ij}^3 + \sum_{j>j'} \mu_j [\nabla (\boldsymbol{\mu}_{j'} \cdot \mathbf{r}_{j'j}/r_{j'j}^3)] + \tfrac{1}{2} \sum_j \alpha_j |\mathbf{E}_j|^2, \tag{11}$$

where the first term comes from the interaction of the charges i with the dipoles j, the second term from the interaction of the dipoles j and j', and the third term from the energy that must be spent in distorting the electron cloud of the atom to create the induced dipoles. Using eqns (7) and (8) we have more simply

$$V_{\mathrm{ind}}^{\mathrm{E}} = - \tfrac{1}{2} \sum_{i,j} Q_i (\boldsymbol{\mu}_j \mathbf{r}_{ij}/r_{ij}^3), \tag{12}$$

where $\boldsymbol{\mu}_j = \alpha_j \sum_i Q_i \mathbf{r}_{ij}/(r_{ij}^3 \epsilon(r))$ from eqns (7) and (9). The energy calculated by this procedure is not very sensitive to the precise value of α, since for a larger value of α, $\epsilon(r)$ also becomes larger.

The electric field calculated with eqns (9) and (11) provides a good approximation to the field $\mathbf{E}_j$ calculated by the rigorous self-consistent procedure of eqns (7) and (8), both for the stabilization of isolated charges and for the interaction between charges inside a protein. As will be seen in the next sections, our approach provides a new insight into the nature of the dielectric effect due to the atoms of the protein. A previous attempt to evaluate the dielectric effect considered only the polarization of groups between the two charges (Hopfinger, 1974), but did not deal with the very important contribution from groups around the charges.

The dielectric effect due to electrostatic interactions with water molecules around the protein (in region III of the complete system, see Fig. 2) is treated differently. In this case the electric field from charges in regions I and II orientates the permanent dipole moments of the water molecules. Most other studies of the dielectric effect of water have been concerned with the actual local electric field on each water molecule inside water solution (see for example, Kirkwood, 1939), and do not provide a practical way to calculate the orientation of a water dipole near charged groups. Our approach is based on an empirical determination of the average orientation of the water dipoles and on an explicit evaluation of the energy of interaction between these oriented dipoles and the charged groups. As a first step, the water molecules, treated as point dipoles, are placed on a cubic lattice with a unit cell dimension of 3 Å. The protein is then placed inside this lattice, and all water molecules that are less than 3 Å from the protein are removed. Next, the first shell of water molecules is rearranged to get a better packing around the protein. This lattice rearrangement is done by minimizing an artificial energy function based on a 10–6 Lennard–Jones-type potential between the water molecules and the atoms on the protein surface. The electric field from the atomic partial charges of the protein interact with these point dipoles of the water molecules causing them to become partially aligned with the electric field. On average, the component of the water dipole so oriented is given by the well-known Langevin formula.

$$\boldsymbol{\mu}_k = \mu_0\{\coth(x_k) - 1/x_k\}\mathbf{E}_k/|\mathbf{E}_k| \tag{13}$$

$x_k = C\mu_0|\mathbf{E}_k|/kT$ and μ_0 is related to the permanent dipole moment of water, k is Boltzmann's constant, T is the absolute temperature, and C reflects the resistance to orientation of the water molecules due, for example, to hydrogen bonds between water molecules.

The component, $\boldsymbol{\mu}_k$ of the water dipole oriented in the direction of applied electric field depends on the strength of that field, while the field in turn depends on both the external field from protein partial charges and from the dipole components $\boldsymbol{\mu}_k$. Once again an iterative self-consistent procedure was used to solve for the $\boldsymbol{\mu}_k$ and $\mathbf{E}_k$, and the energy of the system calculated as

$$V^{\mathrm{W}}_{\mathrm{ind}} = -\sum_{i,k} Q_i\mathbf{r}_{ik}\boldsymbol{\mu}_k/r^3_{ik} + \sum_{k>k'} \boldsymbol{\mu}_k\{\nabla(\boldsymbol{\mu}_{k'}\cdot\mathbf{r}_{k'k}/r^3_{k'k})\}. \tag{14}$$

In this case no energy must be spent in creating the dipoles, as they already exist, and the work done in orienting them is already included in eqn (13). Suitable values of μ_0 and C were found by trying to fit the solvation energies in water of ions with different radii and net charges of 1, 2 and 3 electrons, respectively. The value found for μ_0 of 1·9 D is slightly larger than in the *in vacuo* dipole moment of water (1·69 D). The value found for C of 0·5 might reflect the intrinsic resistance of water dipoles to orientation by an applied field: for a given electric field and temperature the average component of the dipole in the direction of the field is smaller than for dipoles that do not interact (i.e. $C = 1$). A more detailed description of this model will be given elsewhere.

The electric field found by the self-consistent procedure can be used as before (eqn 10)) to find the variation of the effective dielectric $\epsilon(r)$ with distance from a central ion. The calculations showed that $\epsilon(r)$ increases from a value of about 3·5 for the water dipoles closest to the central charge ($r = 3$ Å) to about 8 at $r = 7$ Å. Due possibly to the finite size of our water lattice, which only extended 10 Å from the central charge, $\epsilon(r)$ dropped for $r > 7$ Å, instead of increasing to the value of the macroscopic dielectric constant of water (80). As a compromise we took

$$\epsilon(r) = 1 + r. \tag{15}$$

When this dielectric is used to calculate the local field on the water dipoles, no self-consistent procedure need be used and the calculation is much more efficient. When calculating the local electric field due to a group of atoms, r was taken as the shortest distance between the water point dipole and the group (eqn (9)).

The dipoles induced on the enzyme atoms in region II and the water dipoles oriented in region III stabilize any charged groups in region I through the energy terms given in eqns (12) and (14). They also effect the quantum mechanical calculation in the same way as did the charges on the enzyme: the electrostatic potential due to these dipoles must be added to the diagonal elements of the F matrix.

$$(\Delta F^{ii}_{\mu\mu})_{\mathrm{ind}} = \sum_j \boldsymbol{\mu}_j\mathbf{r}_{ij}/r^3_{ij} + \sum_k \boldsymbol{\mu}_k\mathbf{r}_{ik}/r^3_{ik} \tag{16}$$

(j in region II k in region III).

(c) *Changing the substrate conformation*

Conformational changes of the substrate were generated by minimizing the complete energy function described in the previous sections with respect to all the Cartesian coordinates of the substrate. The enzyme was usually kept fixed at its initial conformation, although selected side chains were moved on occasion. A quadratically convergent minimization method VA09A, by R. Fletcher and taken from the Harwell Subroutine Library was used to minimize the energy by shifting zones of atoms, typically containing all the atoms of two saccharide residues (see Levitt, 1974*b*). Having the first derivative of the energy in analytical form makes the complete relaxation of the system under any set of conditions computationally efficient. Without such relaxation, a few unnecessarily close contacts could give a very misleading energy value.

3. Results

The commonly accepted catalytic mechanism for the cleavage of hexasaccharides by lysozyme (Phillips, 1966; Blake *et al.*, 1967; Vernon, 1967; Dahlquist *et al.*, 1969; Chipman & Sharon, 1969; Imoto *et al.*, 1972) can be summarized as follows. Upon binding of the substrate to the active site, the sugar ring in subsite D is distorted towards a half-chair conformation, thereby weakening the bond between atoms $C_{(1)}$ and $O_{(4)}$ of residues D and E. This bond is further weakened when Glu35 acts as a general acid catalyst, donating a proton to atom $O_{(4)}$. Cleavage of the $C_{(1)}$—$O_{(4)}$ bond leads to the formation of a transition state in which $C_{(1)}$ and $O_{(5)}$ of the D ring are positively charged (oxocarbonium ion) and consequently stabilized in the enzyme active site by the negative charge of Asp52. The charge sharing between $C_{(1)}$ and $O_{(5)}$ tends to make the $C_{(1)}$—$O_{(5)}$ torsion angle planar adding intramolecular stabilization to the half-chair conformation. Once the two saccharides cleaved off leave the active site, the carbonium ion reacts with a water molecule and then it too leaves the active site. An alternative mechanism involves collapse of the ion pair to form a covalent bond between Asp52 and $C_{(1)}$ (Lowe, 1967) and is not considered here.

Vernon (1967) and Doonan *et al.* (1970) have summarized this mechanism clearly and consider three factors that might contribute to the overall catalytic effect: (1) acid catalysis in the protonation of the glycosyl oxygen by Glu35, (2) ring strain of substrate residue D induced by the substrate–enzyme binding, and (3) stabilization of the carbonium ion by interaction between the developing positive charge and the negative charge on Asp52. Here we use the computational scheme described above to estimate the relative importance of factors (2) and (3). The study of acid catalysis in the proton transfer step is left for a subsequent study.

The starting co-ordinates used here are set DLN taken from Levitt (1974*b*); they are based on the extensively refined lysozyme co-ordinate set ER5D (Diamond, 1974; Levitt, 1974*a*) and the rough wire-model co-ordinates measured by Phillips and co-workers (Phillips 1970; and personal communication). All the hydrogen atoms were added in standard positions. We concentrate here on the relative stability and equilibrium conformations of the initial enzyme–substrate complex and the carbonium ion intermediate. As the real activated complex in lysozyme catalysis possesses significant carbonium ion character (Dahlquist *et al.*, 1967), we believe that a detailed study of these two states should give an idea of the relative importance different factors play in reducing the activation energy of the reaction. In the ground state of the bound substrate Glu35 is protonated, and in the carbonium ion intermediate the proton has been transferred from Glu35 to $O_{(4)}$ and the glycosidic bond "cleaved" by setting the $C_{(1)}$—$O_{(4)}$ bond length to 3 Å.

(a) *Strain effects in the ground state*

The proposed role of strain in the catalytic mechanism of lysozyme is based on X-ray crystallographic determination of the atomic structure of the complex between lysozyme and the tri-NAG (tri-*N*-acetyl glucosamine) inhibitor and on model building of the hexa-NAG substrate in the active site of the enzyme (Blake *et al.*, 1967). The model building indicates that the saccharide residue in the D subsite makes bad contacts with the enzyme and that these contacts can be relieved by deformation of the ring towards a half-chair. More recently, Ford *et al.* (1974) studied the crystal structure of a complex between lysozyme and the transition state analogue tetrasaccharide-lactone (TACL) and found that the TACL was bound in subsites A to D. The lactone ring was close to the position of the fourth NAG residue built into the D subsite by Blake *et al.* (1967), but the hydroxymethyl group was less deep in the active-site cleft. They also showed that the most likely conformation of the lactone ring, which has sp^2 geometry at atom $C_{(1)}$, was the "sofa" rather than the half-chair. On the other hand, energy calculations (Levitt, 1972,1974*b*), which allowed the enzyme–substrate complex to relax by energy minimization, showed that the substrate as model-built by Blake *et al.* (1967) could be accommodated in the active site with a full-chair D residue conformation and without any strain, even when the enzyme was held completely rigid. Here we re-examine this conclusion and, in particular, check whether the presence of ionized Asp52 induces some deformation of the chair conformation; this inductive effect can only be examined using quantum mechanical energy calculations.

The calculations of the interconversion from chair to a planar $C_{(1)}$—$O_{(5)}$ system were done by the method of adiabatic mapping (Lifson & Warshel, 1968; Warshel & Karplus, 1974; Levitt, 1974*b*). In this approach one internal co-ordinate (here the $C_{(1)}$—$O_{(5)}$ torsional angle) is held close to a given value by an external constraint, while the energy of the system is minimized with respect to all Cartesian co-ordinates (here the co-ordinates of saccharide D). The stepwise variation of the value to which torsional angle is constrained results in different values of the energy after the minimization, so providing the adiabatic potential energy as a function of the constrained variable. When such mapping is done using convergent minimization techniques, it provides the lowest energy path between the initial and the final point. In the case of the sugar ring, the lowest energy path leads from the full-chair at $\phi_{C(1)-O(5)} = -60°$ to the so called "sofa" conformation at $\phi_{C(1)-O(5)} = 0°$ (Ford *et al.*, 1974), rather than to the half-chair conformation. As seen from the results of this calculation (Fig. 4), the sofa conformation is about 6 kcal/mol less stable than the full-chair conformation. Steric interactions between the enzyme and substrate reduce this difference very slightly to about 5 kcal/mol, as the strong steric interactions are relaxed by small movements of the strongly interacting atoms. On the other hand, electrostatic interactions with the enzyme increase this difference to about 8 kcal/mol. Asp52, which is ionized and negatively charged, interacts favourably with the positively charged $H_{(1)}$ atom (Fig. 1) of the D sugar ring, and the Asp $O^{\delta 2} \ldots H_{(1)}$ separation increases in going from chair to sofa conformations (from 2·32 Å to 3·67 Å). The present calculation differs from the previous one (Levitt, 1974*b*) in that now part of the system is treated quantum mechanically and all the hydrogen atoms are included. Nevertheless, the results of the two calculations are the same: there is no torsional deformation in the initial stage of the reaction.

Our results seem to contradict those obtained from binding studies (Chipman & Sharon, 1969), which estimate that the binding in subsite D is unfavourable relative to other subsites by about 3 kcal/mol, and that indicates steric strain. However, as was pointed out before (Levitt, 1974*b*), the unfavourable binding free energy observed for subsite D may reflect the energy of expelling tightly bound water molecules from

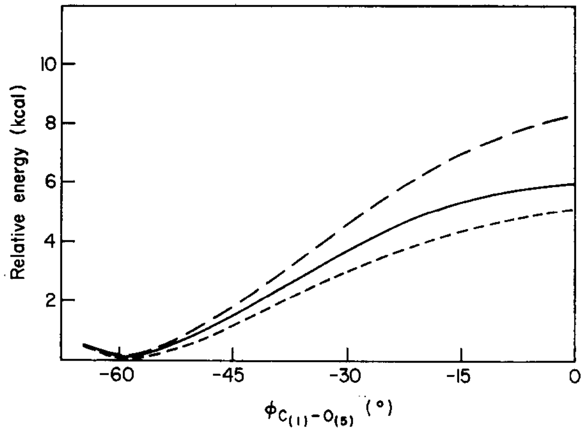

FIG. 4. The adiabatic potential energy curve for the chair–sofa conversion of the sugar ring bound in sub-site D. At each set value of $\phi_{C(1)-O(5)}$, the atoms of sugar ring D are allowed to move to minimize the energy, but the $\phi_{C(1)-O(5)}$ angle is constrained to remain at the set value. The solid line is the intermolecular potential energy of the substrate when not bound to the enzyme (*in vacuo*). The short dash line is the potential that includes steric interaction with the enzyme. The long dash line is the potential that includes both electrostatic and steric interactions with the enzyme (Asp52 is in the negatively charged ionized state). $\phi_{C(1)-O(5)} = -60°$ and $\phi_{C(1)-O(5)} = 0°$ correspond to the chair and the sofa conformations, respectively.

Asp52. In any case, the assumptions (Chipman & Sharon, 1969) about the actual mode of binding of the tetra-NAG inhibitor used to estimate the low binding energy of subsite D have recently been questioned (Levitt, 1974*b*). Comparison of the binding energies of tetrasaccharides that must bind in subsites A to D showed that most of the apparent unfavourable binding energy associated with subsite D comes from the presence of an *N*-acetyl muramic acid saccharide in subsite D.

(b) *Equilibrium conformation of the carbonium ion*

In the second stage of the calculation we evaluate the conformation of the D sugar ring in the carbonium ion transition state and estimate the influence of stain on that conformation. The adiabatic potential energy surface for this region is presented in Figure 5. This time the energy is calculated as a function of two constrained variables, the torsional angle about the $C_{(1)}$—$O_{(5)}$ bond, $\phi_{C(1)-O(5)}$, and the out of plane angle of the

$$O_{(5)}\text{—}C_{(1)}\begin{matrix}\diagup H_{(1)} \\ \diagdown C_{(2)}\end{matrix}$$

system, χ. The carbon $C_{(1)}$ has a planar sp^2 geometry for $\chi = 0°$ and a tetrahedral sp^3 geometry for $\chi = 54°$. Relaxation of the carbonium ion from the initial ground state conformation to the stable planar sofa conformation results in about 45 kcal/mol stabilization. While this value may be too high, we obtained a similar value (40 kcal/mol) by minimizing the corresponding MINDO/2 potential energy (Dewar & Haselbach, 1970; McIver & Komornicki, 1971). A simple classical energy calculation which uses reasonable quadratic force constants of planar hydrocarbons gives a value of ~35 kcal/mol. As is seen from Figure 5, the initial energy relaxation is

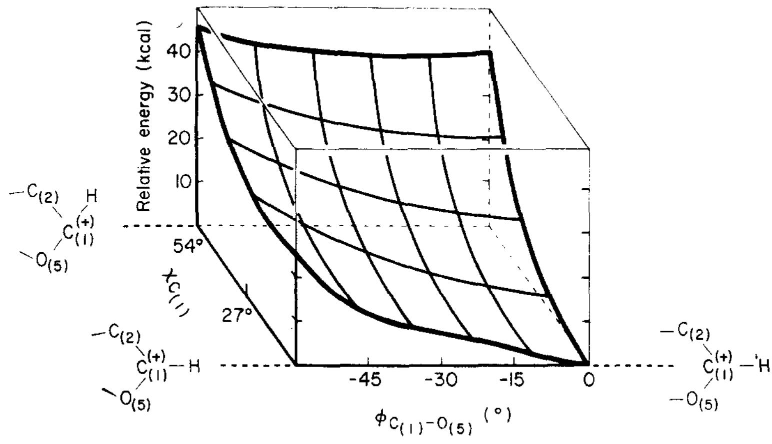

FIG. 5. The calculated adiabatic potential energy surface for the transition between the ground state and carbonium ion conformation of the D sugar ring. The proton $H^{\varepsilon 2}$ has been transferred from Glu35 to $O_{(4)}$ and the $C_{(1)}$—$O_{(4)}$ bond stretched to 3 Å. $\phi_{C(1)-O(5)}$ is the $C_{(5)}$—$O_{(5)}$—$C_{(1)}$—$C_{(2}$ torsional angle; χ is the out-of-plane angle of the

$$O_{(5)}\text{—}C_{(1)}\Big\langle\begin{matrix}H_{(1)}\\C_{(2)}\end{matrix}$$

system. The ground state conformation has sp^2 geometry at $C_{(1)}$ ($\chi = 54°$) and a full chair ring conformation ($\phi_{C(1)-O(5)} = -60°$). The stable conformation of the carbonium ion has planar sp^2 geometry at $C_{(1)}$ ($\chi = 0°$) and a sofa ring conformation ($\phi_{C(1)-O(5)} = 0°$).

mainly due to the motion of the hydrogen $H_{(1)}$ into the $O_{(5)}C_{(1)}C_{(2)}$ plane, rather than to the torsional angle movement from the full-chair to the sofa ring conformation.

Another aspect of the carbonium ion deformation was studied by calculating the adiabatic potential energy map for different $C_{(1)}$—$O_{(5)}$ torsional angles and different $C_{(1)}$—$O_{(4)}$ bond lengths (Fig. 6). At each pair of values of $\phi_{C(1)-O(5)}$ and $b_{C(1)-O(4)}$ the D sugar ring was allowed to relax. In order to keep the sugar in the protonated form always, the $H^{\varepsilon 2}$—$O_{(4)}$ bond length was constrained to be 1·1 Å. It is clear that the $\phi_{C(1)-O(5)}$ torsional potential changes its nature as the glycosidic bond is lengthened; the minimum is initially at the chair conformation ($\phi_{C(1)-O(5)} = 60°$) for a normal $C_{(1)}$—$O_{(4)}$ bond but moves to the sofa conformation ($\phi_{C(1)-O(5)} = 0°$) for a cleaved bond ($b_{C(1)-O(4)} = 3{\cdot}0$ Å). Note that the energy of the ground state after stretching the glycosidic bond ($\phi_{C(1)-O(5)} = 60°$; $b_{C(1)-O(4)} = 3$ Å) is about 24 kcal/mol. This value is less than the maximum value obtained in Figure 5, as the $C_{(1)}$ carbon has relaxed to the planar sp^2 geometry ($\chi = 0°$). The steric effect of the enzyme on this potential surface is the same in both the ground state and in the carbonium ion transition state.

The calculated geometries of the D sugar ring in the ground state and the carbonium ion transition state are given in Tables 1 and 2 and shown in Figure 7. The ground state has lowest energy in the full-chair conformation, whereas the transition state has the sofa conformation with sp^2 geometry at $C_{(1)}$ with atom $H_{(1)}$ in the $O_{(5)}C_{(1)}C_{(2)}$ plane (see Fig. 6). This conformational change is generated spontaneously by energy minimization once the $C_{(1)}$—$O_{(4)}$ bond is stretched to 3 Å. It is important to note that almost the same geometrical change is obtained when the calculation is performed without any steric interaction with the enzyme. The atomic co-ordinates of the D

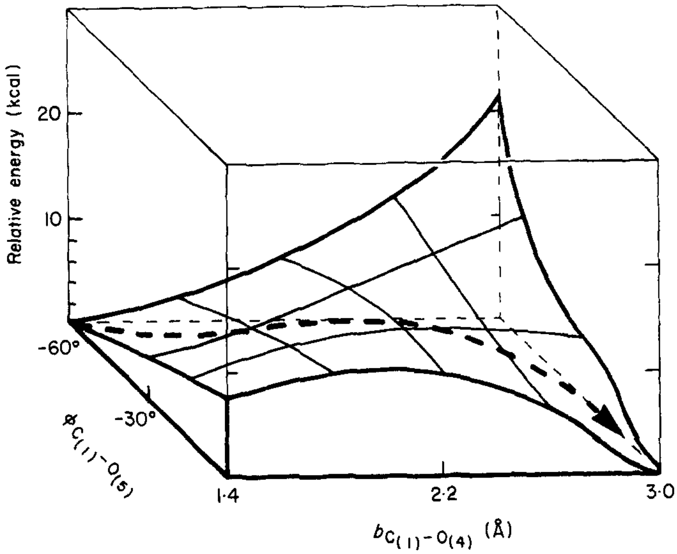

FIG. 6. The adiabatic potential energy surface of sugar D for torsion around the $C_{(1)}$—$O_{(5)}$ torsional angle and stretching of the $C_{(1)}$—$O_{(4)}$ glycosidic bond. The proton $H^{\varepsilon 2}$ has been transferred from Glu35 to $O_{(4)}$, and the $H^{\varepsilon 2}$—$O_{(4)}$ distance constrained to remain at 1·1 Å, so that the sugar remains protonated.

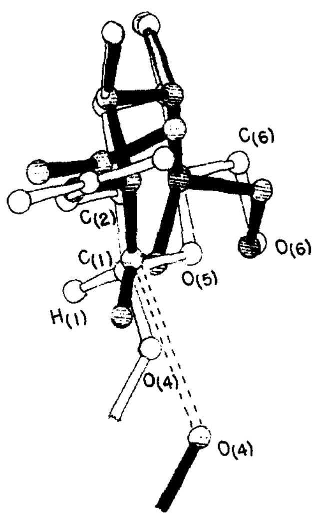

FIG. 7. A comparison of calculated equilibrium conformations of the ground state full-chair (clear bonds) and the carbonium ion intermediate (solid bonds). The calculations were done with the complete enzyme–substrate system, which is omitted from the drawing for greater clarity. Note how the N-acetyl and $C_{(6)}$—$O_{(6)}$ groups move when the sugar ring becomes more planar.

sugar residue in our calculated carbonium ion intermediate differ by 1·1 Å root-mean-square from the co-ordinates of tetrasaccharide-lactone determined by Ford *et al.* (1974). The net shift is —0·43 Å along x (our co-ordinates move more into the active site) and 0·47 Å along y (more to the right of the cleft as viewed in Figs 3 and 4 of Ford *et al.*, 1974). The most striking difference is that atom $O_{(6)}$ is about 2 Å closer to Gln57 in our co-ordinates, yet still within hydrogen-bonding distance of Glu35.

TABLE 1

Calculated equilibrium geometries of the ground state and the carbonium ion intermediate

Internal co-ordinate	Ground state Chair	Ground state Sofa	Carbonium ion Sofa
Bond lengths (Å)			
$C_{(1)}$—$C_{(2)}$	1·57	1·54	1·42 (1·52)†
$C_{(2)}$—$C_{(3)}$	1·53	1·52	1·49 (1·51)
$C_{(3)}$—$C_{(4)}$	1·57	1·57	1·55 (1·51)
$C_{(4)}$—$C_{(5)}$	1·56	1·55	1·56 (1·53)
$C_{(5)}$—$O_{(5)}$	1·48	1·50	1·50 (1·52)
$C_{(1)}$—$O_{(5)}$	1·46	1·45	1·37 (1·32)
Bond angles (°)			
$C_{(2)}$—$C_{(1)}$—$O_{(5)}$	107	119	128 (121)
$C_{(2)}$—$C_{(1)}$—$O_{(4)}$	111	110	— (121)
$C_{(1)}$—$C_{(2)}$—$C_{(3)}$	113	110	109 (100)
$C_{(2)}$—$C_{(3)}$—$C_{(4)}$	108	107	106 (108)
$C_{(3)}$—$C_{(4)}$—$C_{(5)}$	112	112	113 (112)
$C_{(4)}$—$C_{(5)}$—$O_{(5)}$	108	112	115 (114)
$C_{(5)}$—$O_{(5)}$—$C_{(1)}$	117	120	117 (124)
Torsional angles (°)			
$C_{(5)}$—$O_{(5)}$—$C_{(1)}$—$C_{(2)}$	−60	0	−1 (1)
$O_{(5)}$—$C_{(1)}$—$C_{(2)}$—$C_{(3)}$	57	35	31 (30)
$C_{(1)}$—$C_{(2)}$—$C_{(3)}$—$C_{(4)}$	−55	−62	−58 (−59)
$C_{(2)}$—$C_{(3)}$—$C_{(4)}$—$C_{(5)}$	53	46	60 (62)
$C_{(3)}$—$C_{(4)}$—$C_{(5)}$—$O_{(5)}$	−53	−24	−31 (−31)
$C_{(4)}$—$C_{(5)}$—$O_{(5)}$—$C_{(1)}$	58	3	−1 (0)

† Values in parentheses refer to sofa conformation of *N*-acetyl gluconolactone as obtained from the analysis of the electron density map for site D by Ford *et al.* (1974).

(c) *Charge stabilization and dielectric effect*

In the generally accepted reaction mechanism of lysozyme, Asp52 must be in the ionized form and carry a negative charge. However, it is not obvious that Asp52 can be ionized in our quantum mechanical scheme (or indeed in any other scheme) unless it can be stabilized by some form of solvation. When no dielectric is used in our calculations and the enzyme is treated as if in a vacuum, ionization of Asp52 by removing a proton increases the energy of the system by hundreds of kcal/mol, and could never happen. As acidic groups on protein surfaces certainly are ionized under normal conditions, this result is clearly wrong. The cause of this energy imbalance can best be illustrated by considering the thermochemical cycle of ionizing a water molecule in water solution (see Table 3). While the estimated energy required for the ionic dissociation of an O—H bond in vacuum is about 390 kcal/mol, we gain about 270 kcal from the complete solvation of H^+ and about 105 kcal/mol from the complete solvation of the OH^- ion. The whole cycle, therefore, costs only 15 kcal, which is the observed energy for dissociation of a water molecule in water solution. The dissociation of an acid follows a similar pattern with considerable stabilization by solvation of the ionized species. As the pK_a of an acidic group on a protein is similar to that when free in water, it must also be considerably stabilized by solvation.

TABLE 2

Cartesian co-ordinates† of the equilibrium conformations of the D sugar residue and parts of Glu35 and Asp52

Atom	Ground state chair			Carbonium ion sofa		
D Sugar						
$C_{(1)}$	8·38	24·96	20·91	8·34	25·35	21·26
$H(C_{(1)})$	8·93	24·13	20·77	8·92	25·48	20·42
$C_{(2)}$	8·79	25·89	22·11	8·65	26·14	22·40
$H(C_{(2)})$	8·28	26·73	22·02	8·01	27·02	22·30
$N_{(2)}$	10·27	26·27	22·12	10·12	26·52	22·36
$H(N_{(2)})$	10·88	25·64	21·59	10·81	25·95	21·79
$C_{(7)}$	10·68	27·52	22·27	10·53	27·77	22·65
$O_{(7)}$	9·93	28·52	22·40	9·75	28·75	22·83
$C_{(8)}$	12·19	27·70	22·24	12·04	28·05	22·60
$H(C_{(8)})$	12·63	26·93	21·60	12·47	27·62	21·70
$H(C_{(8)})'$	12·43	28·68	21·83	12·22	29·13	22·59
$H(C_{(8)})''$	12·58	27·62	23·25	12·52	27·63	23·49
$C_{(3)}$	8·44	25·30	23·47	8·34	25·35	23·63
$H(C_{(3)})$	9·03	24·42	23·67	8·97	24·46	23·72
$O_{(3)}$	8·71	26·29	24·48	8·57	26·21	24·72
$H(O_{(3)})$	8·33	26·05	25·40	8·23	25·84	25·62
$C_{(4)}$	6·90	24·97	23·47	6·84	24·99	23·55
$H(C_{(4)})$	6·32	25·89	23·44	6·24	25·89	23·57
$C_{(5)}$	6·48	24·08	22·27	6·49	24·16	22·35
$H(C_{(5)})$	6·93	23·08	22·36	6·53	23·07	22·57
$C_{(6)}$	4·98	23·96	22·22	5·03	24·44	21·93
$H(C_{(6)})$	4·52	24·91	22·43	4·95	25·40	21·44
$H(C_{(6)})'$	4·61	23·21	22·91	4·36	24·42	22·80
$O_{(6)}$	4·63	23·58	20·90	4·60	23·44	21·03
$H(O_{(6)})$	5·43	23·19	20·43	4·62	22·49	21·41
$O_{(5)}$	6·96	24·71	21·02	7·34	24·40	21·13
Glu35‡						
C^{γ}	4·15	23·19	17·45			
C^{δ}	4·70	24·51	17·98			
$O^{\varepsilon 1}$	5·95	24·62	18·04			
$O^{\varepsilon 2}$	3·98	25·33	18·59			
Asp52						
C^{β}	8·85	19·81	21·44			
C^{γ}	8·89	20·97	22·42			
$O^{\delta 1}$	8·99	20·79	23·65			
$O^{\delta 2}$	8·88	22·15	21·98			

† The co-ordinates are given in the enzyme co-ordinate system.
‡ The co-ordinates of Glu35 and Asp52 are the same for both sugar conformations.

With our dielectric model, the calculations do indeed show that the ionized Asp52 is stabilized greatly by interactions with the enzyme and the bound substrate in the ground state. This extra stabilization amounts to ~58 kcal/mol (close to the 65 kcal/mol solvation energy expected for a negative ion of the size of COO^-, see Friedman & Krishann, 1973). An interesting result of our work is that about 29 kcal/mol of the stabilization energy comes from the atomic polarizability in region II of Figure 3, while 29 kcal/mol comes from hydrogen bonds of Asp to the protein, and only 2 kcal/mol from interactions with the surrounding water. (The individual contributions

TABLE 3

The thermochemical cycle of water dissociation†

Process	Enthalpy change (kcal/mol)
$H_2O + H_2O \rightarrow H^+ + OH^- + H_2O$	+390
$H_2O + H^+ + OH^- \rightarrow H_3O^+ + OH^-$	−155
Water + $H_3O^+ \rightarrow (H_3O^+)_{liq}$	−115
Water + $OH^- \rightarrow (OH^-)_{liq}$	−105
$(H_2O)_{liq} + (H_2O)_{liq} \rightarrow (H_3O^+)_{liq} + (OH^-)_{liq}$	+15

† Estimated from the results of De Paz *et al.* (1970) and from the references given by Hepler & Wooley (1973).

total 60 kcal/mol, but the internal energy of the system increases by 2 kcal/mol (see Table 4), giving the net stabilization of 58 kcal/mol.) This indicates that the pK values of polar groups on a protein are similar to those in water due to the good interactions with dipoles induced in the protein.†

Table 4 gives the energy contributions for the minimum energy conformations of the substrate under various conditions. The difference in energy of the substrate when in vacuum and when bound to the protein (−30 kcal/mol) cannot be compared to the measured subsite binding energy. The binding energy of the substrate to the protein should be evaluated relative to the energy of the substrate inside water. While it would be possible to use our water model to estimate the energy of the substrate in water, we did not perform this calculation in the present work.

When Asp52 is ionized in the ground state conformation there is an apparent extra stabilization of 58 kcal/mol. In the calculations of the ionization of Asp52 we did not consider the complete cycle in which an O—H bond is broken and the H^+ subsequently stabilized by interaction with the solvent. When these factors are considered, as was done for the simple case of water dissociation (Table 3), the energy change on ionization is only a few kcal/mol. Because these neglected energy contributions are constant for all the configurations of the substrate considered here, it is still possible to calculate the electrostatic stabilization due to Asp52 in the ionized form. With the carbonium ion in the planar sofa conformation, ionization of Asp52 gives net stabilization of 9 kcal/mol (Table 4, rows 11 and 12). The system is stabilized as soon as the proton is transferred from Glu35 to $O_{(4)}$, even without cleavage of the glycosidic bond, as the ionized Asp52 interacts with the protonated saccharide. The electrostatic stabilization due to the ionization of Glu35 cannot be estimated separately, as this residue is an integral part of the region studied quantum mechanically; the stabilization of the carbonium by the ionized Glu35 tends to compensate for the increase in energy when the proton is transferred to $O_{(4)}$.

† The concept of a strong stabilization of an ion by a low dielectric medium can be best demonstrated by the somewhat oversimplified Born model (Born, 1920) for ion solvation. In this model the solvation free energy V_{sol} is given (in kcal/mol) by

$$V_{sol} = \frac{166}{r_0}\left(1 - \frac{1}{\epsilon}\right), \tag{17}$$

where r (which is given in Å) is the effective radius of the ion. When in a medium with $\epsilon = 2$, one gets about half as much solvation as when in water ($\epsilon = 80$).

TABLE 4

Relative energy contribution of different configurations of the D sugar ring

Configuration		Condition† Enzyme	Asp^{COO^-}	V_{ind}	V_{QQ}	Relative energy contribution‡ V_{vdw}	V_{nonb}	V_{intra}	V_{tot}	$V_{tot\ MINDO}$
Ground state	1	–	–	0	0	0	0	0	0	
chair	2	+	–	−10	−8	−14	−32	2	−30	
	3	+	+	−10(−41)§	−8(−37)	−14	−32(−92)	2(4)	−30	−30
Ground state	4	–	–	0	0	0	0	6	6	
sofa	5	+	–	−12	−6	−15	−33	8	−25	
	6	+	+	−15	0	−15	−30	7	−23	−24
Carbonium ion	7	–	–	0	0	0	0	93	93	
chair	8	+	–	−15	−12	−14	−41	87	46	
	9	+	+	−9	−40	−14	−63	100	37	38
Carbonium ion	10	–	–	0	0	0	0	48	48	
sofa with	11	+	–	−20	−7	−15	−42	47	1	
planar C_1 system	12	+	+	−15	−33	−15	−63	55	−8	−2

† The heading *Enzyme* indicates whether the energy includes non-bonded interactions with the enzyme and with induced dipoles in the enzyme and water regions. The heading Asp^{COO^-} indicates whether Asp52 is ionized or not.

‡ V_{ind} is the interaction of the charges in region (I) with the induced dipoles in region (II) (the enzyme) and with the oriented dipoles in region (III) (the surrounding solvent). V_{QQ} is the charge–charge interaction between the atoms of the D residue of the substrate and the enzyme atoms (excluding the charge–charge interaction between the substrate and Glu35, which is included in the quantum energy). V_{vdw} is the van der Waals interaction between the D residue atoms and the enzyme. V_{nonb} is the sum of the non-bonded contributions V_{ind}, V_{QQ}, and V_{vdw}. V_{intra} is the internal (quantum mechanical) energy of the substrate and the carboxyl group of Glu35. V_{tot} is the total energy of the substrate atoms in subsite D relative to the ground state chair form in vacuum. $V_{tot\ MINDO}$ is the V_{tot} where the quantum mechanical energy is evaluated by the MINDO/2 potential surface (in the carbonium ion case, the MINDO/2 Configuration Interaction correction was estimated by performing separate calculations for the carbonium ion and the leaving group).

§ All entries with Asp52 ionized COO^- have been corrected to allow for the decrease in energy due to the interaction of Asp52 with the rest of the system (with the ground state chair conformation for the D ring). This stabilization of Asp52 on becoming ionized is 58 kcal/mol, made up of $\Delta V_{ind} = 31$ kcal/mol, $\Delta V_{QQ} = 29$ kcal/mol, $\Delta V_{intra} = -2$ kcal/mol as the energy of actually removing a proton from Asp52 was not included (see text for a fuller discussion). The uncorrected values are given in parentheses in row 3.

To understand more fully the electrostatic interaction between Asp52 and the protonated D saccharide, we left Glu35 out of the quantum calculation in the neutral state and protonated the D saccharide by binding an extra hydrogen to $O_{(4)}$. The results of this calculation are shown in Figure 8. The electrostatic interaction energy of Asp52 with the rest of the system is -61 kcal/mol when the saccharide is unprotonated. The corresponding energy of the protonated saccharide is -40 kcal/mol when Asp52 is not ionized. When both groups are charged and interact, the electrostatic energy is -120 kcal/mol, a net stabilization of only 19 kcal/mol relative to the situation when these charged groups do not interact with one another. It is interesting that the actual charge–charge interaction V_{QQ} between both charged groups is -100 kcal/mol, the value expected from the interaction of a pair of unit charges of opposite sign 3·3 Å apart *in vacuo*, which gives an apparent dielectric constant of $100/19 \approx 5$. Here the dielectric effect arises not by attenuation of the interaction between the charged groups, but rather by the more favourable interaction of the separated groups with the medium. In the different situation when a pair of positively charged groups come together, the charge–charge interaction, V_{QQ}, is large and positive. This energy increase is compensated, however, by the strongly favourable interaction, V_{ind}, of the doubly charged cluster with the polarizable medium (V_{ind} varies approximately as $(Q_{\mathrm{total}})^2$). In general, groups with greater and more concentrated net charge, or higher charge asymmetry (large dipole moment), interact more favourably with the dielectric (V_{ind}), but these same groups have a more unfavourable internal electrostatic energy from interaction between charges within the group (V_{QQ}). This balance illustrates how the dielectric medium influences energies when it is included explicitly as in this work, and gives a clearer physical picture than when the dielectric is considered to attenuate the charge–charge interactions.

Table 5 gives the atomic partial charges of the atoms treated quantum mechanically for both the chair and sofa forms. The charge on $H_{(1)}$ is more positive in the chair conformation than in the sofa conformation, reflecting the closer contact and stronger electrostatic interaction with negatively charged Asp52. These charge changes cause

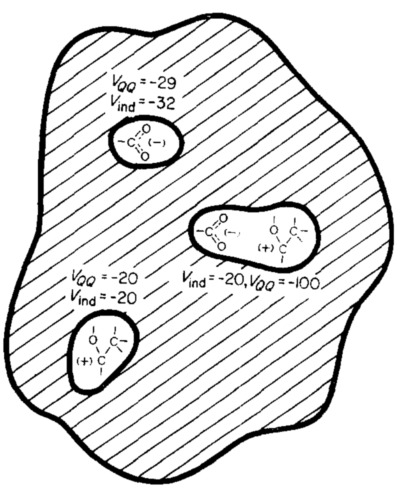

FIG. 8. A schematic representation of the balance between solvation energy and the formal electrostatic interaction that leads to the dielectric effect.

TABLE 5

Calculated charge distribution of the quantum atoms in different conformations

Residue	Atom	Conformation Ground state chair	Ground state sofa	Carbonium ion chair	Carbonium ion sofa
D ring	$C_{(1)}$	0·22 (0·36)	0·33 (0·38)	0·60 (0·57)	0·56 (0·63)
	$H_{(1)}$	0·06 (−0·05)	−0·01 (−0·06)	0·20 (0·08)	0·10 (0·12)
	$C_{(2)}$	0·00 (−0·32)	−0·04 (−0·33)	0·15 (−0·16)	0·12 (−0·18)
	N_2^*	−0·11 (0·09)	−0·10 (0·09)	−0·05 (0·16)	−0·06 (0·18)
	H_2	0·01 (0·05)	−0·01 (0·05)	0·01 (0·05)	0·02 (0·08)
	C_3^*	0·01 (0·05)	0·02 (0·05)	0·03 (0·04)	0·02 (0·04)
	C_5^*	0·18 (0·18)	0·16 (0·18)	0·28 (0·27)	0·27 (0·29)
	$O_{(5)}$	−0·24 (−0·24)	−0·18 (−0·24)	−0·27 (−0·31)	−0·04 (−0·19)
	$O_{(4)}$	−0·28 (−0·28)	−0·27 (−0·29)	−0·47 (−0·48)	−0·38 (−0·47)
	C^*	0·14 (0·15)	0·14 (0·16)	0·19 (0·19)	0·16 (0·18)
Glu35	$O^{\varepsilon 1}$	−0·61 (−0·53)	−0·61 (−0·53)	−0·78 (−0·62)	−0·80 (−0·71)
	$O^{\varepsilon 2}$	−0·29 (−0·27)	−0·29 (−0·27)	−0·77 (−0·63)	−0·76 (−0·77)
	$H^{\varepsilon 2}$	0·18 (0·16)	0·18 (0·16)	0·22 (0·21)	0·17 (0·21)
	C^{δ}	0·78 (0·64)	0·79 (0·64)	0·79 (0·67)	0·79 (0·66)
	C^{γ}	−0·06 (0·01)	−0·07 (0·01)	−0·16 (−0·06)	−0·17 (−0·09)

† The charges given in parentheses are for the corresponding calculation *in vacuo* without any charge–charge, induced dipole, or van der Waals interaction with the enzyme.

changes in the energy contributions: V_{ind} is −10 kcal/mol in the ground state chair and −15 kcal/mol in the ground state sofa, while the corresponding values for V_{QQ} are −8 kcal/mol and 0 kcal/mol (see Table 4). Because of the energy compensation described above, the net change is smaller, and the chair is only 3 kcal/mol more more stable. The *in vacuo* charges sometimes differ significantly from those calculated when bound to the enzyme. In one case, Glu35 is more completely ionized when there is dielectric stabilization. In another case, the *N*-acetyl group ($N_{(2)}^*$ and $H_{(2)}$) becomes more negative due to a hydrogen bond to $N^{\delta 1}$ of Asn46. These differences are even more marked for some of the intermediate conformations generated as the $O_{(4)}$—$C_{(4)}$ bond is first cleaved. Clearly, the electrostatic field of the induced dipoles and partial charges on the enzyme and the oriented dipoles in the surrounding solvent affect not only the energy but also the charge distribution of the quantum mechanical system.

4. Conclusion

We have presented a general method for a detailed study of enzymic reactions. The method treats the whole enzyme–substrate–solvent system and includes all the energetic factors that might contribute to the reaction. These factors include the energy of bond rearrangement and charge redistribution during the reaction, the steric and the electrostatic interaction between the enzyme and the substrate, and the dielectric effect due to polarization of both the enzyme and the surrounding water molecules. The incorporation of the polarizability of the atoms of the protein into the calculation allows us to reproduce, for the first time, the energetic balance found in hydrogen transfer reactions and to deal properly with electrostatic interactions.

We have applied the method to study the factors that influence the stability of the carbonium ion intermediate in the reaction of lysozyme. It was found that the effect of steric strain is unlikely to be important because of its rapid relaxation by small shifts in the co-ordinates of the substrate and the enzyme. On the other hand, it was found that electrostatic interactions play a very important role in the enzymatic reaction. The total electrostatic stabilization of the carbonium ion relative to vacuum was found to be very large (—40 kcal/mol; see Fig. 8). While this result is theoretically important in that it shows how the polarizability model accounts for the stabilization of ionized groups on the protein, it is not biochemically relevant. Normally the reaction rate and transition state stabilization of the enzyme is compared to that in water so that the relative stabilization of the carbonium ion is much smaller than 40 kcal/mol. Once it is clear that the large solvation effect is due mainly to the polarizability of the protein, the relevant question is how much of the ionized Asp52 contributes to the stabilization (relative to the non-ionized case). Here we find that the presence of Asp52 lowers the energy of the carbonium relative to the ground state by about 9 kcal/mol.

In this work we have concentrated mainly on the relative stability of the ground and transition states of the D ring of the substrate. The detailed energetics and pathway of the hydrogen transfer step (general acid catalysis) from Glu35 is still under study. Preliminary calculations show that it is possible to compare the energetics of proton transfer from Glu35 to that from a water molecule in solution, and so clarify the importance of acid catalysis in each case.

In view of the calculations we feel that the accepted mechanism of lysozyme (Phillips, 1966) has to be modified to emphasize "electrostatic strain" rather than "steric strain". When the substrate binds to lysozyme, part of the binding energy is used to increase the electrostatic energy of Asp52 by displacing tightly bound water molecules rather than to increase the steric strain energy by distorting the D sugar ring. The carbonium ion, which is formed in the transition state, interacts about 9 kcal/mol more favourably with ionized Asp52 than does the ground state. In this way the electrostatic strain is relieved and the reaction rate accelerated by a factor of 3×10^6. In general, it seems much easier to control the reaction rate by electrostatic interactions which vary slowly with distance, than by steric interactions which vary rapidly with distance and can be relaxed by small atomic movements. As the charge distribution on the substrate changes significantly during the lysozyme reaction (Table 5), the enzyme can best control the reaction rate by having charges and effective local dielectric designed to interact well with the charge distribution of the transition state. Although the role of strain in the lysozyme mechanism has been emphasized in the past, Vernon (1967) concluded from a study of model compounds that charge stabilization is likely to be the major factor in the catalysis (see also Doonan *et al.*, 1970).

In this work we have dealt only with the potential energy changes during the reaction and not with the entropy contributions. In our model it is also possible to evaluate explicitly the entropy contributions by calculating the vibrational entropy of the system (Warshel & Lifson, 1970; Warshel & Karplus, 1974). While this contribution was not studied here in detail, a preliminary simplified calculation seems to indicate that the entropy of the bound ground state substrate and the bound carbonium ion transition state are similar.

We feel that the main value of the present scheme is that it allows one to examine consistently all the important factors in enzymic reactions. The usual theoretical quantum mechanical calculations of bond breaking and ionic processes involve energy changes of several hundred kcal/mol. Our introduction of a microscopic dielectric effect gives the energy balance of real chemical and biochemical reactions where the energy changes are of the order of 10 kcal/mol. On this basis, the model can be used to assess the relative importance of different catalytic factors in enzymes relative to reactions in solution.

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